Associative Property

For any three sets A, B and C

$$A \cup (B \cup C) = (A \cup B) \cup C \quad ; \quad A \cap (B \cap C) = (A \cap B) \cap C$$

Distributive Property

For any three sets A, B and C

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \qquad [\text{Intersection over union}]$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \qquad [\text{Union over intersection}]$$

De Morgan's Laws for Set Difference

For any three sets A, B and C

$$A - (B \cup C) = (A - B) \cap (A - C)$$

$$A - (B \cap C) = (A - B) \cup (A - C)$$

De Morgan's Laws for Complementation

Consider an Universal set and A, B are two subsets, then

$$(A \cup B)' = A' \cap B' \quad ; \quad (A \cap B)' = A' \cup B'$$

Cardinality of Sets

If A and B are any two sets, then $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

If A, B and C are three sets, then

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

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